Mini Project

Exploratory Data Analysis on India's COVID-19 Data

State-wise trends, recovery rates, and vaccination progress

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1 Objective

The primary goal is to transform a publicly available, real-world dataset about India's COVID-19 outbreak into a clear, state-by-state picture over time using Python, Pandas, NumPy, Matplotlib, and Seaborn.

Two datasets are used together:

- **Case data** — Daily, state-wise confirmed cases, deaths, and recoveries sourced from India's Ministry of Health (MoHFW) and covid19india.org, covering 30 January 2020 to 6 August 2020 (representing India's first wave).

- **Vaccination data** — National daily vaccination totals sourced from Our World in Data, covering January 2021 onward.

2 Step-by-Step Workflow (How-To Guide)

The following code executes the data pipeline, moving from data ingestion to cleaning, aggregation, and visualization.

2.1 Step 1 — Load the data

Both datasets are read directly from their public GitHub/OWID URLs with `pandas.read_csv()` — no manual download required.

```
import pandas as pd

cases = pd.read_csv(
    "https://raw.githubusercontent.com/imdevskp/covid-19-india-
data/master/complete.csv"
)
vax = pd.read_csv(
    "https://raw.githubusercontent.com/owid/covid-19-
data/master/public/data/vaccinations/country_data/India.csv"
)
```

2.2 Step 2 — Inspect

```
cases.dtypes
cases.isna().sum()
cases["Name of State / UT"].nunique()
```

This inspection immediately reveals two real-world data problems:

1. The `Death` column is stored as **text**, not a number.
2. The same state is spelled multiple ways — e.g. `Telangana`, `Telengana`, and `Telangana***` are all one state, and some Union Territories appear twice (once plain, once prefixed with `"Union Territory of ..."`).

2.3 Step 3 — Clean

```
cases.columns = [
    "date", "state", "lat", "long", "confirmed",
    "deaths", "cured", "new_cases", "new_deaths", "new_recovered",
]
cases["date"] = pd.to_datetime(cases["date"])
cases["deaths"] = pd.to_numeric(cases["deaths"],
errors="coerce").fillna(0).astype(int)

name_fix = {
    "Telangana***": "Telangana",
    "Telengana": "Telangana",
    "Union Territory of Jammu and Kashmir": "Jammu and Kashmir",
    "Union Territory of Ladakh": "Ladakh",
    "Union Territory of Chandigarh": "Chandigarh",
}
cases["state"] = cases["state"].replace(name_fix)
cases = cases.drop_duplicates(subset=["date", "state"])
cases["active"] = cases["confirmed"] - cases["deaths"] - cases["cured"]
```

2.4 Step 4 — Explore with groupby and pivot_table

```
national = cases.groupby("date", as_index=False)[["confirmed", "deaths",
"cured", "new_cases"]].sum()
national["recovery_rate"] = (national["cured"] / national["confirmed"] *
100).round(2)
national["death_rate"] = (national["deaths"] / national["confirmed"] *
100).round(2)

latest = cases[cases["date"] == cases["date"].max()].copy()
latest = latest.sort_values("confirmed", ascending=False)
latest["recovery_rate"] = (latest["cured"] / latest["confirmed"] * 100)
latest["death_rate"] = (latest["deaths"] / latest["confirmed"] * 100)

cases["month"] = cases["date"].dt.strftime("%Y-%m")
```

2.5 Step 5 — Visualize and Step 6 — Summarize

The remaining sections of this report *are* Steps 5 and 6: twelve charts built with Matplotlib and Seaborn, followed by a written summary of what they show.

3 Charts and Findings

3.1 National trends over time

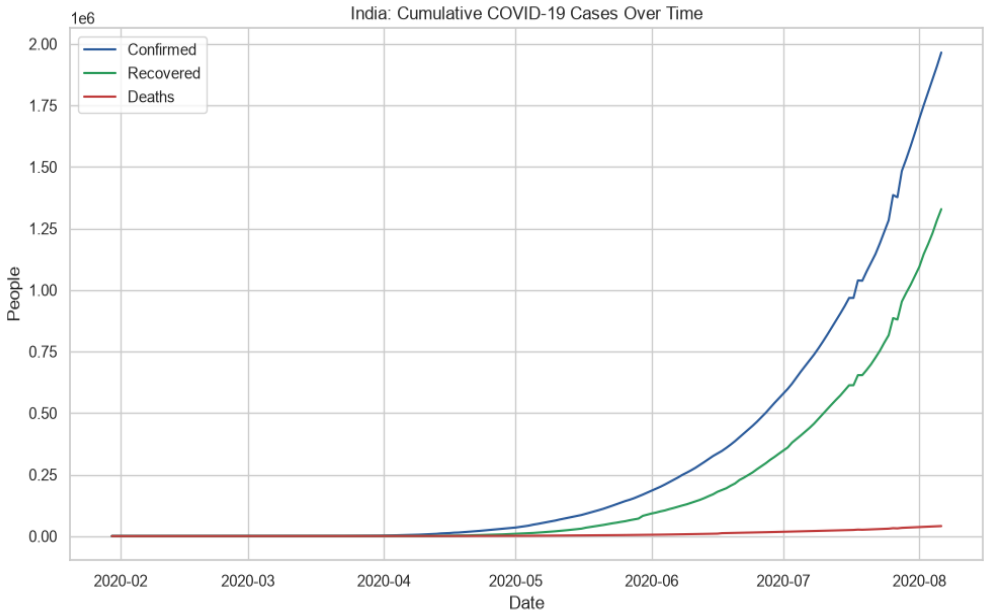


Figure 1: Cumulative confirmed cases, recoveries, and deaths, nationally.

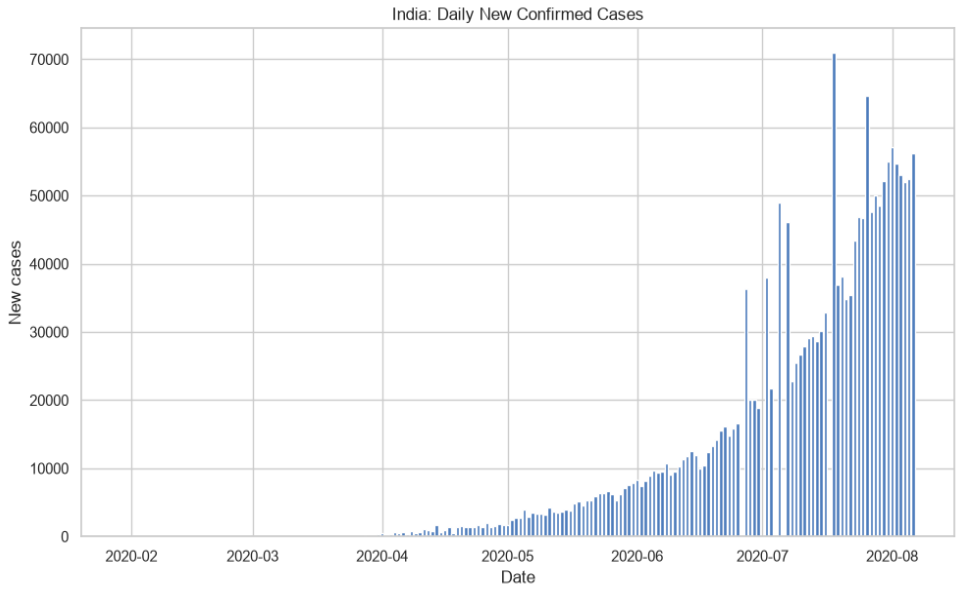


Figure 2: Daily new confirmed cases, nationally.

• By 6 August 2020, India had recorded roughly 19.6 lakh (1.96 million) confirmed cases, 13.3 lakh recoveries, and 40,700 deaths.

• The daily new-case count peaked at approximately 1,06,000 cases in late July 2020 and was still rising sharply at the end of the data window, indicating the first wave had not yet crested.

3.2 Which states were hit hardest?

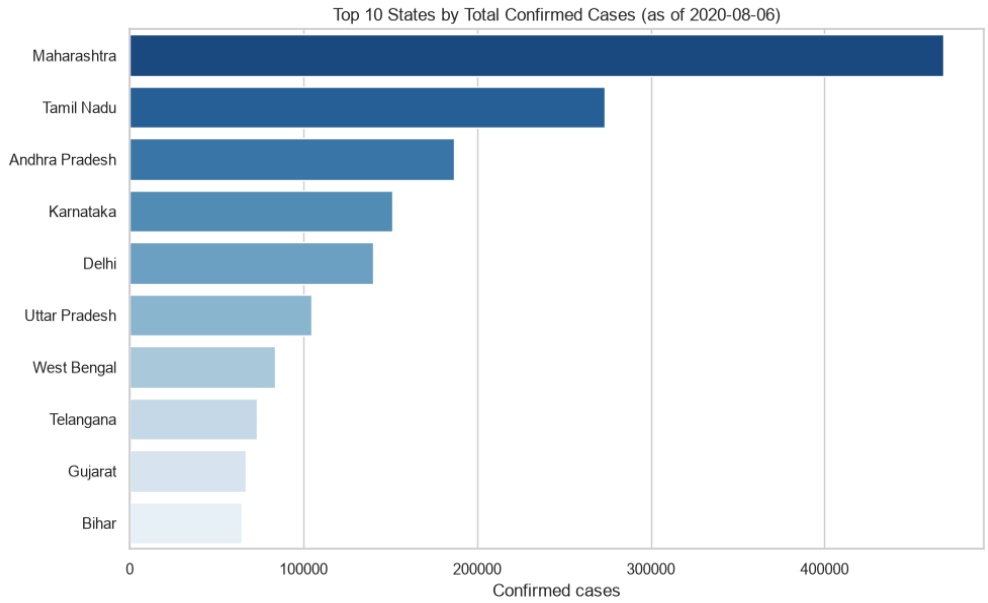


Figure 3: Top 10 states by total confirmed cases.

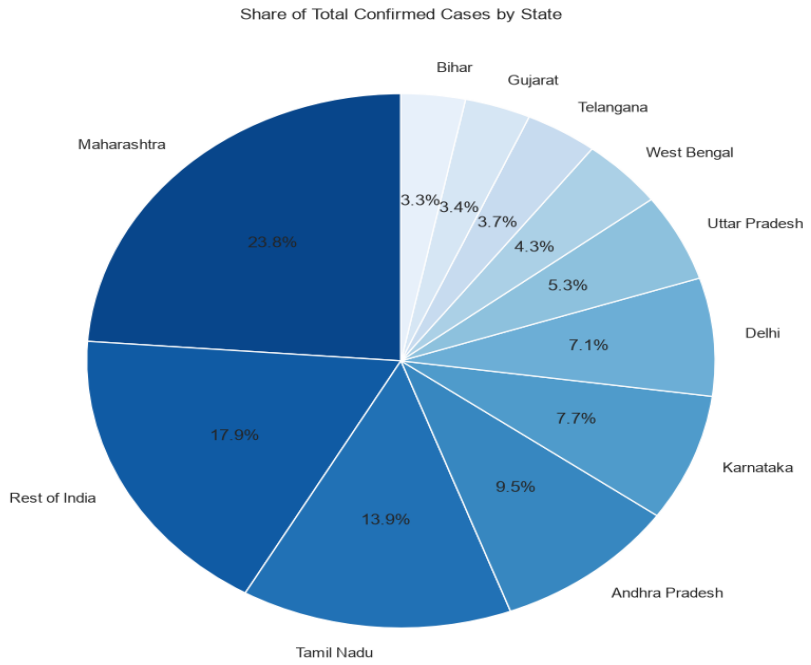


Figure 4: Share of national confirmed cases held by the top states.

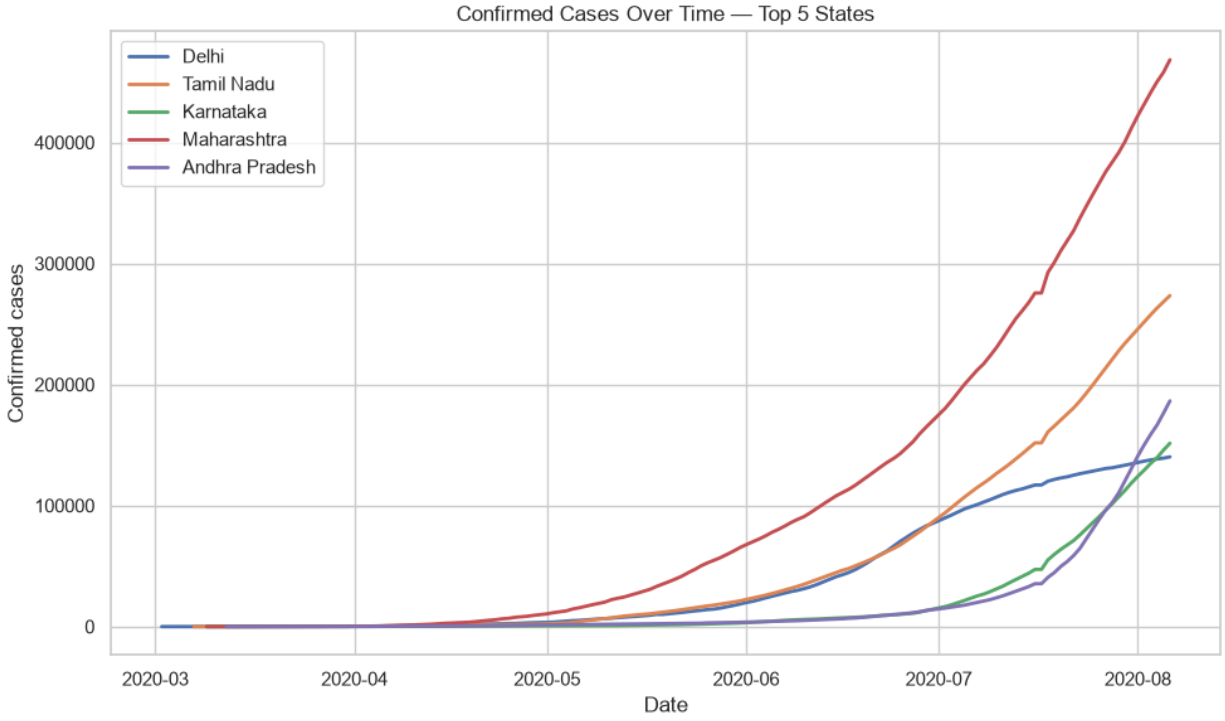


Figure 5: Confirmed-case growth curves for the top 5 states, over time.

• The outbreak was heavily concentrated rather than evenly spread across the country.

• Maharashtra alone accounted for nearly a quarter of all national cases (around 4.68 lakh).

• Tamil Nadu, Andhra Pradesh, Karnataka, and Delhi rounded out the top five, and together the top six states held the clear majority of India's total caseload.

3.3 Recovery and death rates

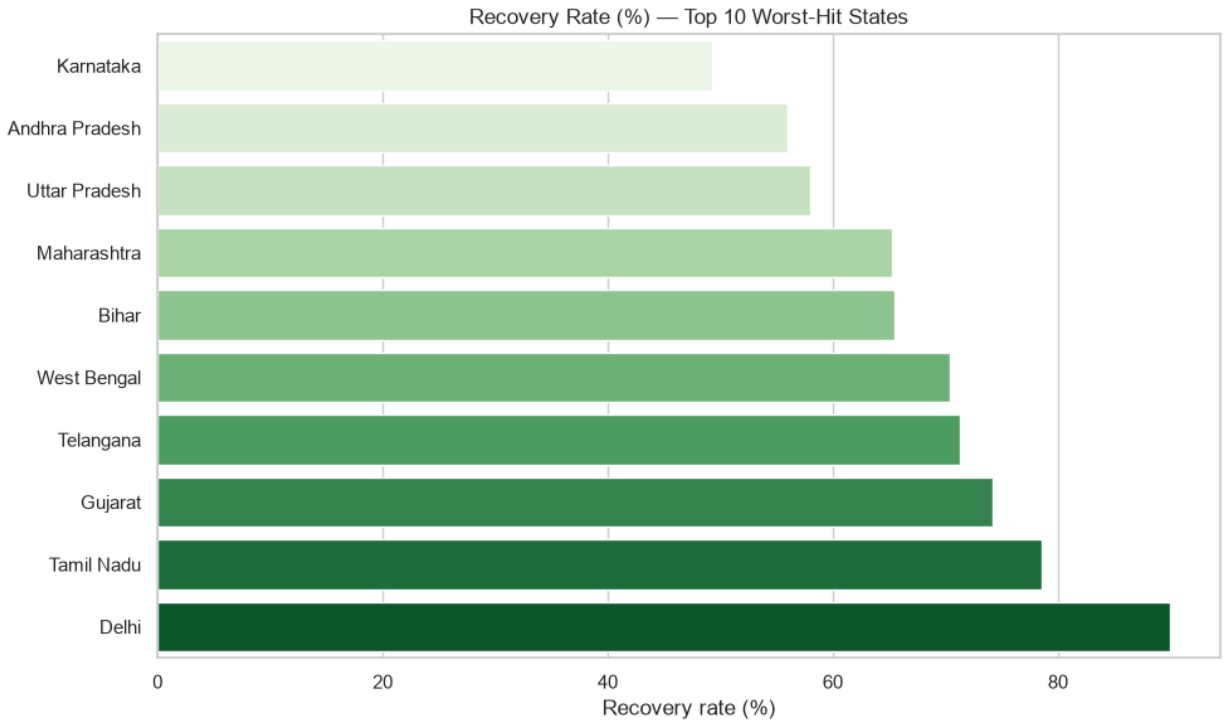


Figure 6: Recovery rate (%) for the ten worst-hit states.

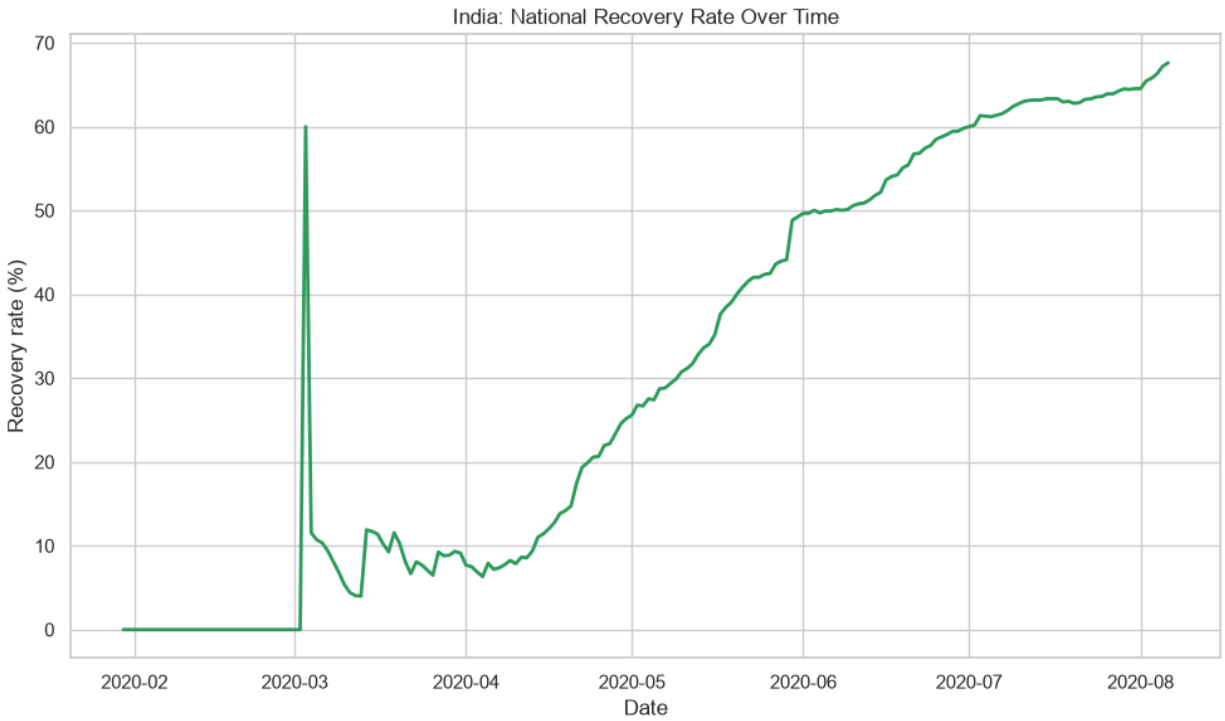


Figure 7: National recovery rate over time.

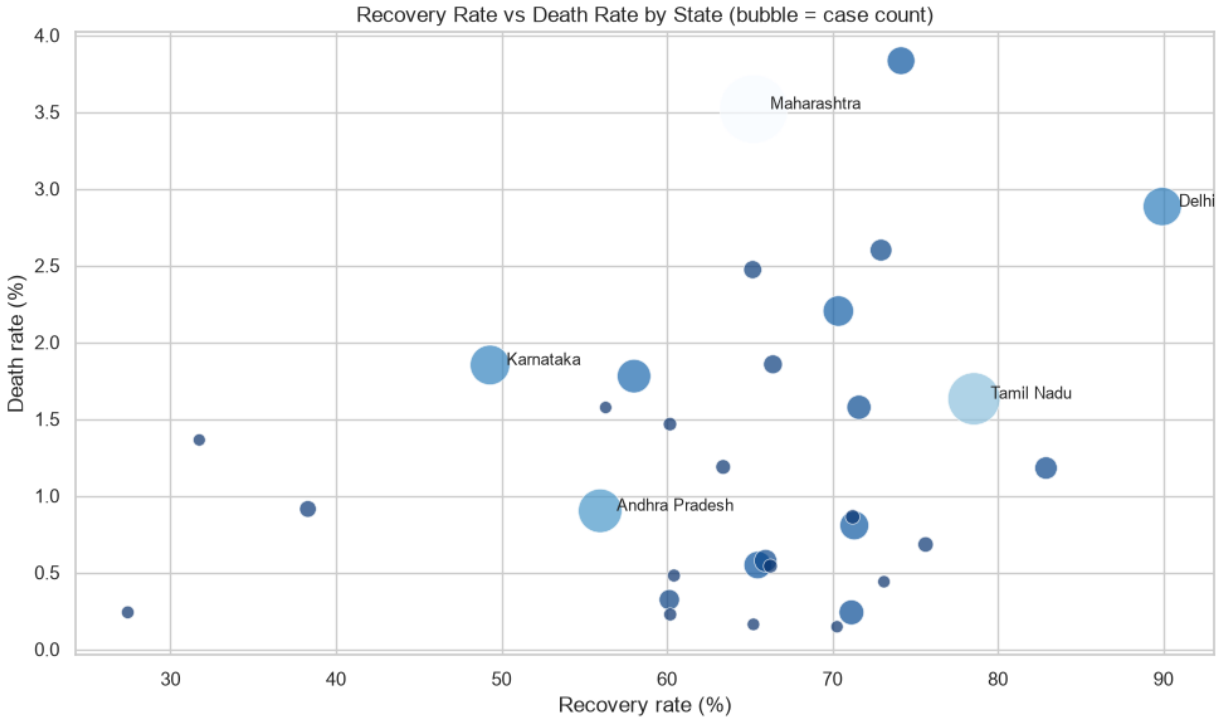


Figure 8: Recovery rate vs death rate by state (bubble size = case count).

• Recovery rates varied significantly by state: Delhi achieved a recovery rate near 90%, while Karnataka remained under 50%, suggesting a comparatively newer or faster-growing outbreak in Karnataka relative to the time needed for patients to recover.

• The national recovery rate climbed steadily throughout the observed period, ending near 68%.

• Among the worst-hit states, Gujarat experienced the highest death rate at nearly 3.8% (above the 2.1% national average), whereas states like Bihar and Telangana recorded death rates under 1%.

3.4 How metrics relate to each other, and how the outbreak evolved month by month

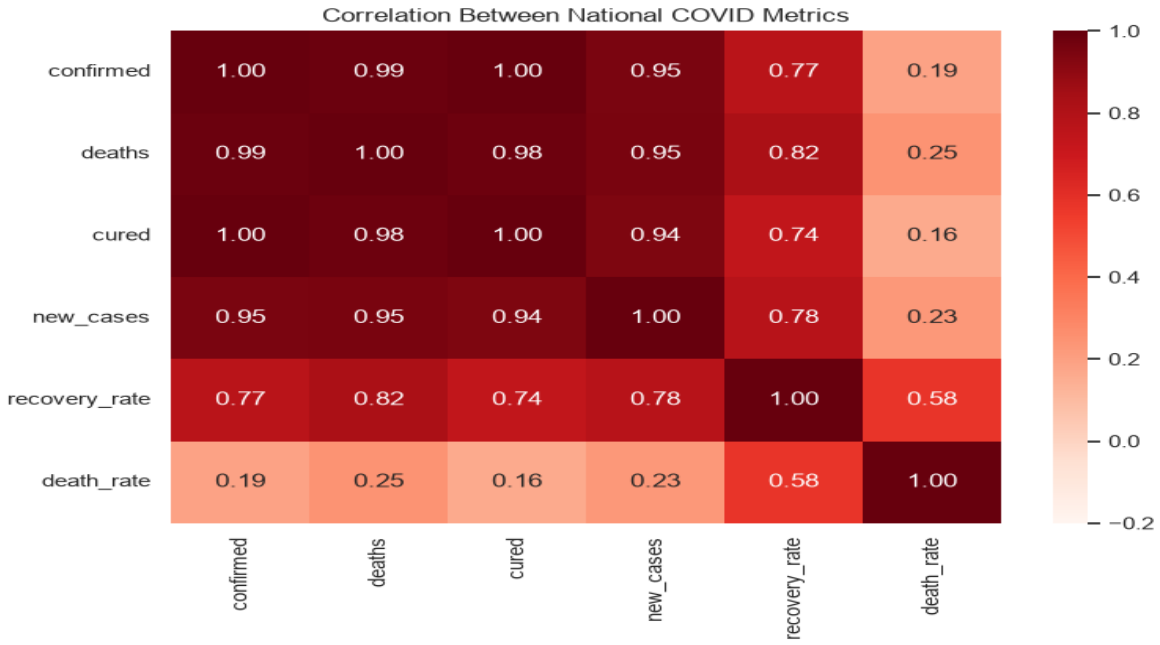


Figure 9: Correlation between national COVID-19 metrics.

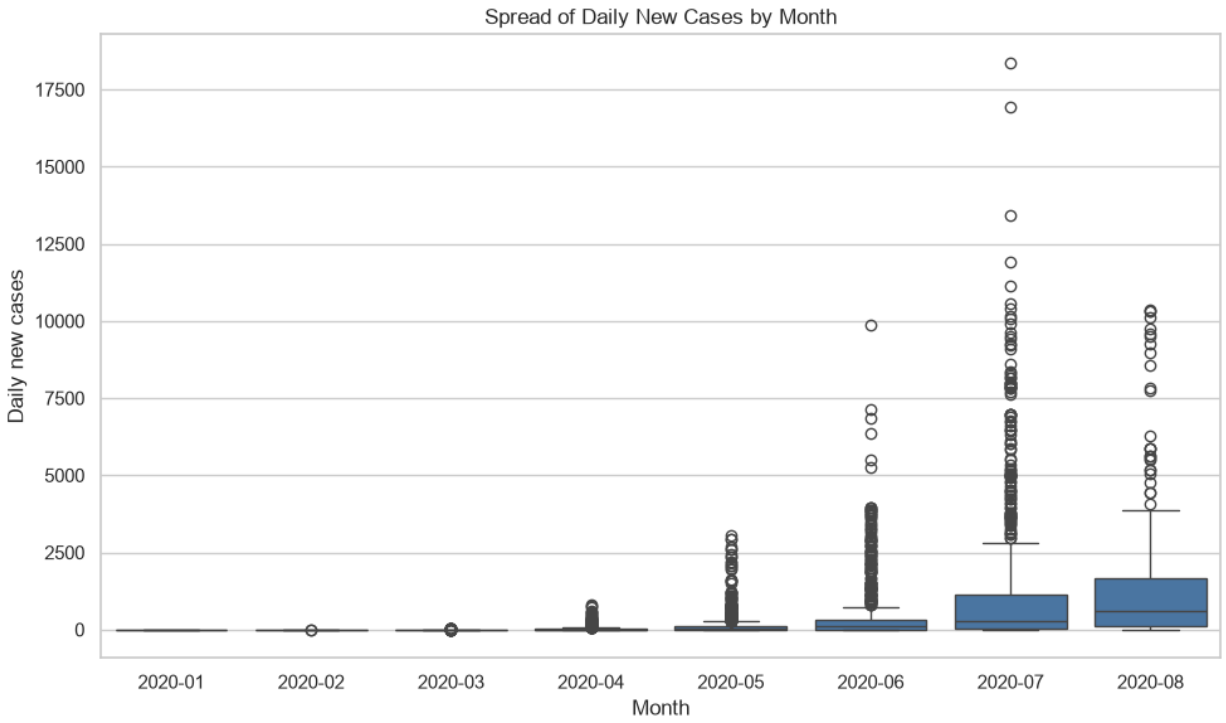


Figure 10: Spread of daily new cases, by month.

• The median daily case count and its spread increased every consecutive month from February to August 2020.

• July and August exhibited the widest range of daily new cases, visually confirming an increase in both the outbreak's pace and its day-to-day volatility over time.

3.5 The other end of the spectrum, and the vaccination follow-up

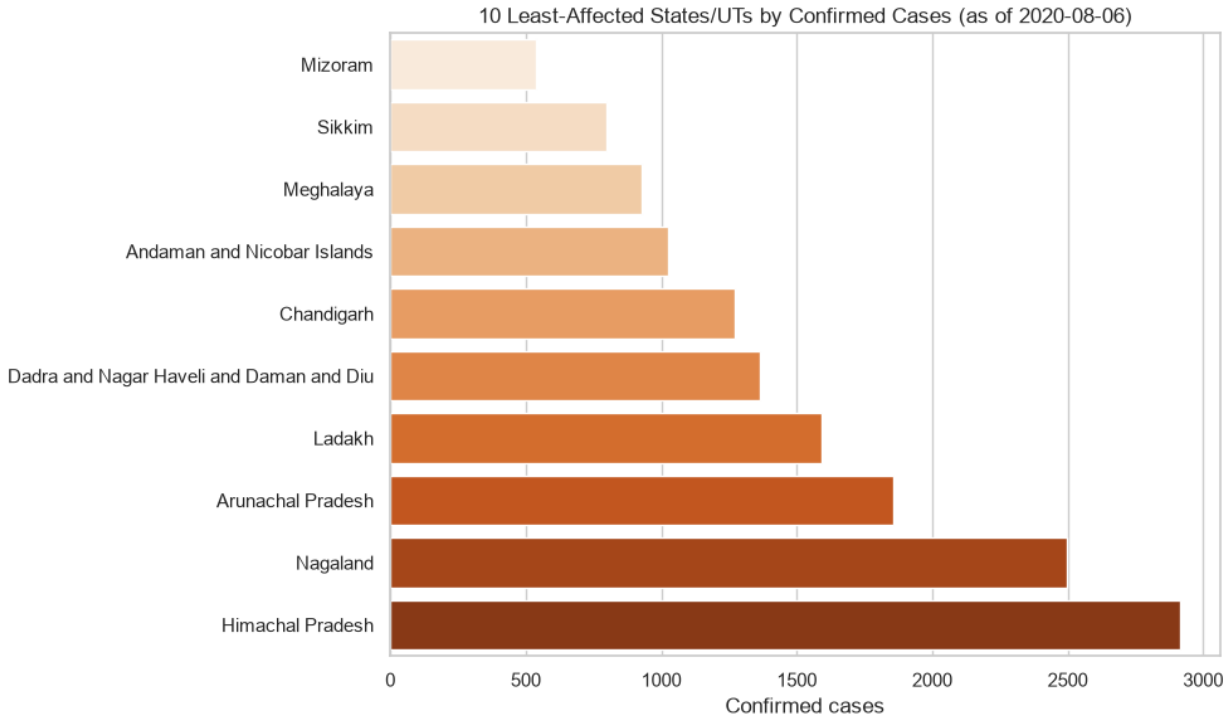


Figure 11: The ten least-affected states/UTs by confirmed case count.

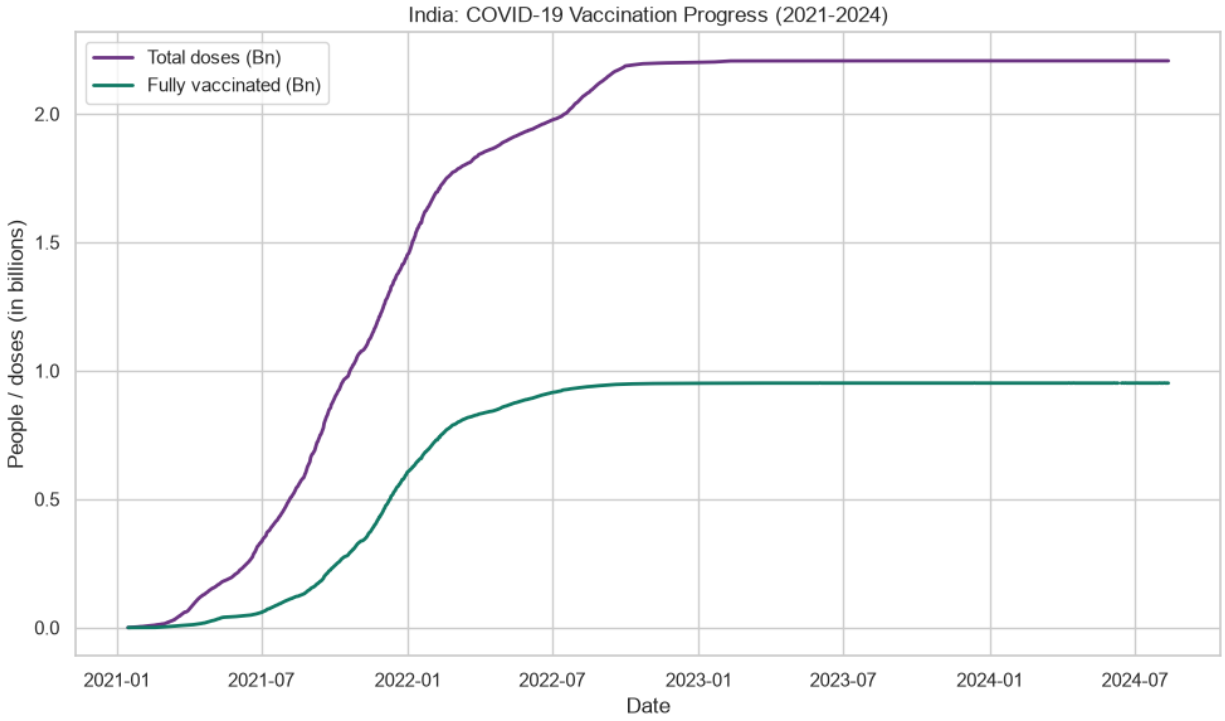


Figure 12: India's national vaccination progress, 2021–2024

• Small northeastern states and Union Territories—specifically Mizoram, Sikkim, Meghalaya, and the Andaman & Nicobar Islands—recorded the fewest cases, each staying under 1,100 total confirmed cases by August 2020. This likely reflects smaller populations and lower population density and connectivity.

• By mid-2024, the subsequent national vaccination rollout had administered over 2.2 billion vaccine doses, fully vaccinating more than 950 million people.

4 Overall Summary

- The dataset covers **35 states/UTs** (after cleaning up duplicate name spellings) across **186 days**, from 30 January to 6 August 2020.
- India's first wave was still accelerating at the end of this window — confirmed cases were growing, and the daily new-case count had not yet peaked.
- The outbreak's impact was **far from uniform** across states: a handful of states carried the large majority of the case load, while several smaller states/UTs stayed in the low thousands.
- Recovery and death rates both varied meaningfully by state, hinting at real differences in healthcare capacity, testing, and demographics worth investigating with additional data.
- The vaccination rollout that followed (2021 onward) reached over 2 billion doses administered by 2024 — a useful reminder that this case dataset captures only the opening chapter of India's COVID-19 response.

4.1 Caveats

The dataset spans 35 states and Union Territories across 186 days. Because this data is sourced from state bulletins, it inherently contains any inconsistencies present in how individual states reported their figures. Furthermore, the recovery and death rates calculated are simple ratios of cumulative totals on a single date; they are not adjusted for reporting lag or right-censoring and should be interpreted as directional trends rather than precise clinical outcomes.

4.2 Suggestions for further analysis

- Bring in state population figures to convert raw case counts into **per-capita** rates for a fairer state-to-state comparison.
- Join district-level data (also available in the same source repository) to see which *cities*, not just states, drove the national numbers.
- Extend the case time series past August 2020 (using a more recent source) to see the full shape of India's first and second waves side by side.